ASSIGNMENT : 14

TITLE : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives : To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory : Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Exercise :

1. Create an ATM program handling invalid withdrawal amount using try-catch.

java14a.java

java14a.java > Language Support for Java(TM) by Red Hat > java14a > main(String[])

```java
import java.util.Scanner;

public class java14a {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int balance = 150000;

        try {
            System.out.print(s: "Enter withdrawal amount: ");
            int amount = sc.nextInt();
            sc.close();

            if (amount > balance) {
                throw new Exception(message: "Insufficient balance.");
            }

            balance = balance - amount;

            System.out.println(x: "Withdrawal Successful.");
            System.out.println("Remaining Balance: " + balance);

        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

OUTPUT :

- ```
  Enter withdrawal amount: 45000
  Withdrawal Successful.
  Remaining Balance: 105000
  PS C:\Users\softlink\Desktop\java code>
  ```

- ```
  Enter withdrawal amount: 200000
  Insufficient balance.
  ```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user.

java14a.java | java14b.java

java14b.java > Java > java14b > main(String[] args)

```java
import java.util.Scanner;

public class java14b {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            System.out.print(s: "Enter product quantity: ");
            int quantity = sc.nextInt();
            sc.close();

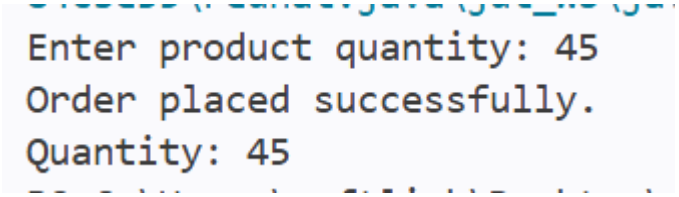
            if (quantity <= 0) {
                throw new Exception(message: "Invalid product quantity.");
            }

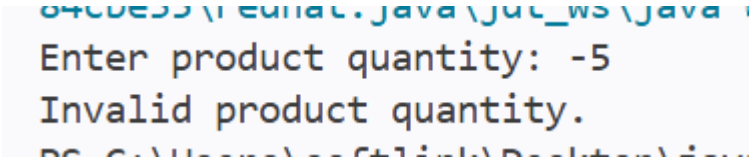
            System.out.println(x: "Order placed successfully.");
            System.out.println("Quantity: " + quantity);

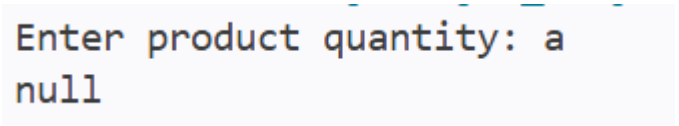
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }


    }
}
```

OUTPUT :

- 

- 

- 

Outcome :

- Understood the concept of exception handling in Java.
- Implemented programs using try-catch, nested try, and multiple catch blocks.
- Learned how exception handling prevents abnormal program termination and improves reliability.

Conclusion :

This assignment demonstrated exception handling using try-catch blocks in Java. Proper exception handling allows programs to manage runtime errors gracefully, ensuring robust, reliable, and user-friendly applications.